

Problem 10

Show that the point

$$z = 1 + i - \alpha - i\beta, \quad \text{for} \\ 0 < \alpha < 1, \quad 0 < \beta < 1, \quad 0 < \alpha + \beta < 1$$

is located in the interior of the triangle with vertices $1, 1 + i, i$.

Solution

We need to prove that z lies strictly inside the triangle with vertices at $1, 1 + i$, and i .

Step 1: Identify the Vertices

The triangle has vertices at:

- $A = 1$ (point $(1, 0)$ in Cartesian coordinates)
- $B = 1 + i$ (point $(1, 1)$)
- $C = i$ (point $(0, 1)$)

This is a right triangle with the right angle at vertex $B = 1 + i$.

Step 2: Express z in Standard Form

$$z = 1 + i - \alpha - i\beta = (1 - \alpha) + i(1 - \beta)$$

In Cartesian coordinates:

$$z = (x, y) \quad \text{where} \quad x = 1 - \alpha, \quad y = 1 - \beta$$

Step 3: Determine Constraints on Coordinates

Given the constraints on α and β :

- From $0 < \alpha < 1$: We get $0 < 1 - \alpha < 1$, so $0 < x < 1$
- From $0 < \beta < 1$: We get $0 < 1 - \beta < 1$, so $0 < y < 1$
- From $0 < \alpha + \beta < 1$: We get $0 < 2 - (x + y) < 1$, so $1 < x + y < 2$

Step 4: Express Interior of Triangle

A point (x, y) is inside the triangle with vertices $(1, 0)$, $(1, 1)$, $(0, 1)$ if and only if it satisfies all three conditions:

Method 1: Using Barycentric Coordinates

A point is inside the triangle if it can be written as a convex combination of the vertices:

$$z = \lambda_1 \cdot 1 + \lambda_2 \cdot (1 + i) + \lambda_3 \cdot i$$

where $\lambda_1, \lambda_2, \lambda_3 > 0$ and $\lambda_1 + \lambda_2 + \lambda_3 = 1$.

Method 2: Using Linear Inequalities

The triangle has three edges. A point is inside if it's on the correct side of all three edges.

The three edges are:

1. From $A = (1, 0)$ to $B = (1, 1)$: vertical line $x = 1$
2. From $B = (1, 1)$ to $C = (0, 1)$: horizontal line $y = 1$
3. From $C = (0, 1)$ to $A = (1, 0)$: line connecting them

Step 5: Check Each Inequality

Condition 1: Right of edge CA

The line from $C = (0, 1)$ to $A = (1, 0)$ has equation:

$$\frac{y - 0}{x - 1} = \frac{1 - 0}{0 - 1} = -1$$
$$y = -(x - 1) = -x + 1$$
$$x + y = 1$$

For a point to be inside (on the right/upper side of this line):

$$x + y > 1$$

From $0 < \alpha + \beta < 1$:

$$(1 - \alpha) + (1 - \beta) = 2 - (\alpha + \beta) > 2 - 1 = 1$$

So $x + y > 1$

Condition 2: Left of edge AB

The edge from $A = (1, 0)$ to $B = (1, 1)$ is the vertical line $x = 1$.

For a point to be inside (to the left):

$$x < 1$$

From $0 < \alpha < 1$:

$$x = 1 - \alpha < 1$$

So $x < 1$

Condition 3: Below edge BC

The edge from $B = (1, 1)$ to $C = (0, 1)$ is the horizontal line $y = 1$.

For a point to be inside (below):

$$y < 1$$

From $0 < \beta < 1$:

$$y = 1 - \beta < 1$$

So $y < 1$

Step 6: Additional Verification

We also need to check that the point is above the x -axis and to the right of the y -axis:

- $x = 1 - \alpha > 0$ (from $\alpha < 1$)
- $y = 1 - \beta > 0$ (from $\beta < 1$)

Step 7: Express Using Barycentric Coordinates

Alternatively, we can write:

$$z = (1 - \alpha) + i(1 - \beta) = (1 - \alpha - \beta) \cdot 1 + \alpha \cdot i + \beta \cdot (1 + i)$$

Wait, let me recalculate:

$$z = 1 + i - \alpha - i\beta = 1(1 - \alpha - \beta) + i \cdot \alpha + (1 + i) \cdot \beta$$

Actually, we can write:

$$z = (1 - \alpha - \beta)(1 + i) + \alpha \cdot 1 + \beta \cdot i$$

No, let me be more careful. We have:

$$z = (1 - \alpha) + i(1 - \beta)$$

This can be written as:

$$z = \alpha \cdot C + \beta \cdot A + (1 - \alpha - \beta) \cdot B$$

where $C = i$, $A = 1$, $B = 1 + i$.

Let's verify:

$$\begin{aligned} \alpha \cdot i + \beta \cdot 1 + (1 - \alpha - \beta)(1 + i) &= \beta + (1 - \alpha - \beta) + i[\alpha + (1 - \alpha - \beta)] \\ &= \beta + 1 - \alpha - \beta + i(1 - \beta) = (1 - \alpha) + i(1 - \beta) = z \end{aligned}$$

Since $\alpha, \beta, (1 - \alpha - \beta) > 0$ and $\alpha + \beta + (1 - \alpha - \beta) = 1$, the point z is a strict convex combination of the vertices, hence inside the triangle.

Final Answer

The point $z = 1 + i - \alpha - i\beta$ lies in the interior of the triangle with vertices $1, 1 + i, i$

Proof summary:

With coordinates $z = (x, y) = (1 - \alpha, 1 - \beta)$, the conditions $0 < \alpha < 1$, $0 < \beta < 1$, $0 < \alpha + \beta < 1$ ensure:

1. $x + y > 1$ (above diagonal edge)
2. $x < 1$ (left of vertical edge)
3. $y < 1$ (below horizontal edge)
4. $x > 0, y > 0$ (in first quadrant)

These conditions define the interior of the triangle.

Alternative proof: $z = \beta \cdot 1 + \alpha \cdot i + (1 - \alpha - \beta)(1 + i)$ is a convex combination of the vertices with positive coefficients, hence lies strictly inside the triangle.